RAG Learning

OCR A A-Level Chemistry — Practical Activity Groups (PAGs)

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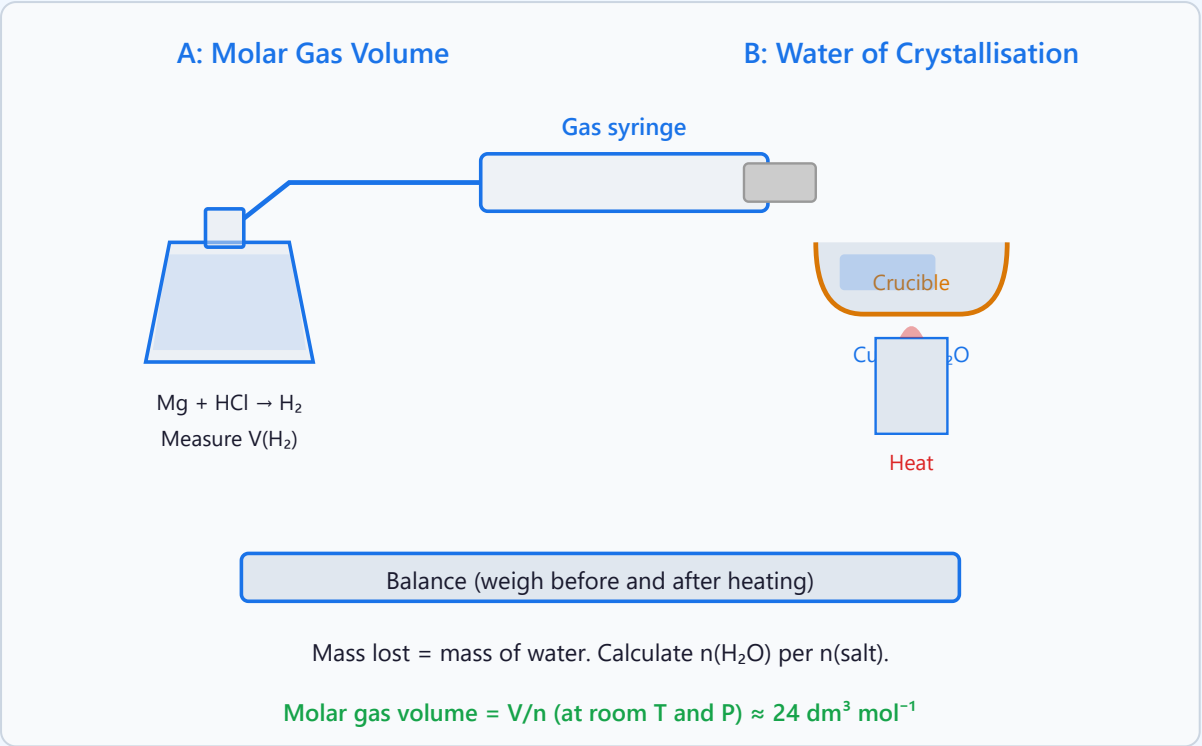
⚗ PAG1: Moles Determination



Apparatus

- **Gas volume:** Conical flask, gas syringe (100 cm^3), delivery tube, balance, measuring cylinder, Mg ribbon, dilute HCl
- **Water of crystallisation:** Crucible, pipe-clay triangle, tripod, Bunsen burner, balance (to 0.01 g), hydrated salt (e.g., $CuSO_4{\cdot}5H_2O$)

Method (A — Molar Gas Volume)

1. Weigh a known length of Mg ribbon accurately.
2. Place excess dilute HCl in a conical flask connected to a gas syringe.
3. Add the Mg ribbon, quickly seal, and collect the H_2 gas.
4. Record the final volume of gas at room temperature and pressure.
5. Calculate: n(Mg) = mass/A_r. Since $Mg + 2HCl \rightarrow MgCl_2 + H_2$, $n(H_2) = n(Mg)$.
6. Molar gas volume = V(gas) / $n(H_2)$.

Method (B — Water of Crystallisation)

1. Weigh a clean, dry crucible. Add hydrated salt and weigh again.

2. Heat gently at first, then more strongly. Allow to cool. Weigh.
3. Reheat, cool, and reweigh until constant mass is achieved (to ensure all water is driven off).
4. Mass of water = mass before - mass after. Mass of anhydrous salt = final mass - crucible mass.
5. Calculate n(water) and n(salt). The ratio gives x in the formula $MX{\cdot}xH_2O$.

Safety

- H_2 gas is flammable — no naked flames near gas syringe method.
- Hot crucible — use tongs; place on heatproof mat.
- Dilute HCl is irritant — goggles.

Analysis

Gas volume: Molar gas volume ≈ 24,000 cm^3 mol^{-1} at room temperature and pressure.

Water of crystallisation: Calculate the integer ratio to find x. For $CuSO_4{\cdot}xH_2O$, expect x = 5.

Error Analysis

Source of Error	Type	Improvement
Gas escaping before bung fitted	Systematic	Seal quickly; use excess acid
Gas syringe friction	Systematic	Tap gently; lubricate
Not heating to constant mass	Systematic	Reheat and reweigh until mass is constant
Salt spitting out of crucible	Systematic	Heat gently at first; use a lid
Reabsorption of water on cooling	Systematic	Cool in a desiccator

Exam Tip: "Heating to constant mass" means heating, cooling, weighing, reheating, and reweighing until the mass doesn't change (difference < 0.01 g). This confirms all water of crystallisation has been removed. Always cool in a desiccator to prevent reabsorption of moisture.

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Worked Example — Water of Crystallisation:

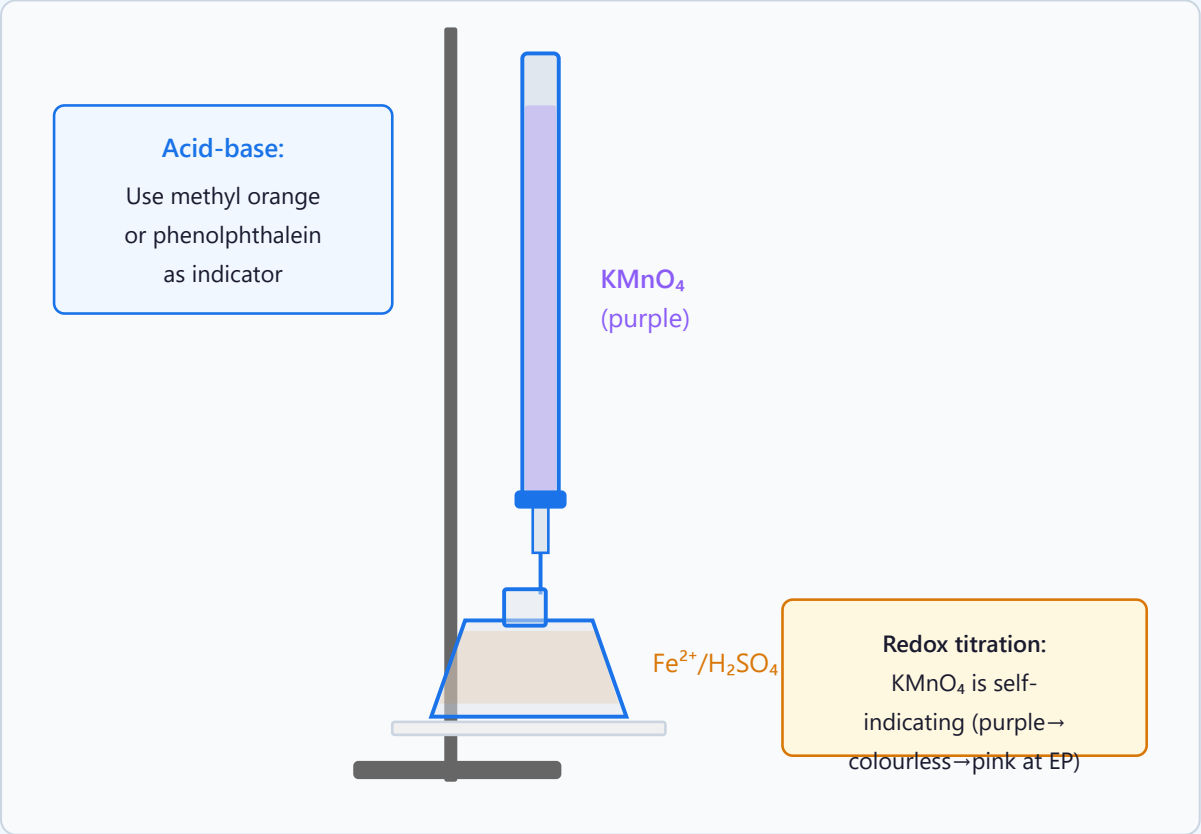
Mass of crucible = 15.20 g. Mass of crucible + hydrated salt = 20.18 g. Mass after heating to constant mass = 18.38 g.

Mass of water lost = 20.18 − 18.38 = 1.80 g → $n(H_2O)$ = 1.80 / 18.0 = 0.100 mol

Mass of anhydrous salt = 18.38 − 15.20 = 3.18 g → $n(CuSO_4)$ = 3.18 / 159.6 = 0.0199 mol

Ratio = 0.100 / 0.0199 = 5.02 ≈ **5**, so formula is **$CuSO_4 \cdot 5H_2O$**

⚗ PAG2: Acid-Base and Redox Titrations



Apparatus

- Burette (50 cm^3), pipette (25.0 cm^3), conical flasks
- White tile, indicators (for acid-base)
- **Acid-base:** Standard acid/alkali solutions
- **Redox:** $KMnO_4$ solution, acidified Fe^{2+} solution (Mohr's salt + H_2SO_4)

Method (Redox — $KMnO_4/Fe^{2+}$)

1. Pipette 25.0 cm^3 of acidified Fe^{2+} solution into a conical flask.
2. Fill the burette with standardised $KMnO_4$ solution.
3. Add $KMnO_4$ slowly, swirling. The purple colour is decolourised as MnO_4^- is reduced to Mn^{2+}.
4. End point: the first permanent pale pink colour that persists for 30 seconds (no indicator needed — $KMnO_4$ is self-indicating).
5. Record the titre. Repeat for concordant results.
6. $MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$ (1:5 ratio).

Safety

- $KMnO_4$ stains skin and is an oxidiser — wear gloves.
- Sulfuric acid is corrosive — goggles.
- Use H_2SO_4 (not HCl or HNO_3) to acidify — HCl would be oxidised by $KMnO_4$, HNO_3 is itself an oxidiser.

Analysis

Calculate: $n(KMnO_4) = c \times V(\text{titre})$. $n(Fe^{2+}) = 5 \times n(KMnO_4)$. Then $[Fe^{2+}] = n/V$.

Error Analysis

Source of Error	Type	Improvement
Overshooting end point	Systematic	Add drop-wise near end; swirl between drops
$KMnO_4$ staining burette (hard to read)	Systematic	Rinse well; read from top of meniscus ($KMnO_4$ is opaque)

Not using H_2SO_4 as the acid	Systematic	Always use sulfuric acid for $KMnO_4$ titrations
Fe^{2+} oxidised by air before titration	Systematic	Use freshly prepared solution; keep acidified

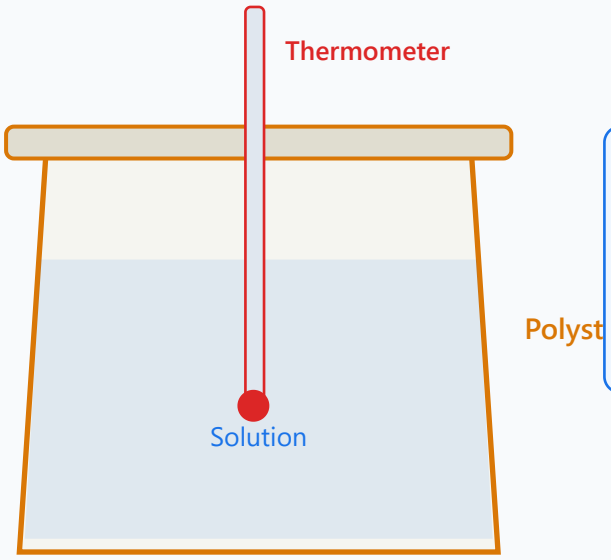
Exam Tip: For $KMnO_4$ titrations, read the burette from the TOP of the meniscus (not the bottom) because the solution is so dark you cannot see through it. No indicator is needed — $KMnO_4$ is self-indicating. The end point is a permanent faint pink colour.

Worked Example — Redox Titration:

25.0 cm^3 of Fe^{2+} solution required 23.60 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$.

$n(KMnO_4) = 0.0200 \times 23.60/1000 = 4.72 \times 10^{-4}$ mol
From equation: $n(Fe^{2+}) = 5 \times n(KMnO_4) = 5 \times 4.72 \times 10^{-4} = 2.36 \times 10^{-3}$ mol
$[Fe^{2+}] = 2.36 \times 10^{-3} / 0.0250 =$ **0.0944 mol dm^{-3}**

⚗ PAG3: Enthalpy Changes by Calorimetry



Polyst

Hess's Law

$\Delta H_1 = \Delta H_2 + \Delta H_3$

Measure ΔH_2 and ΔH_3

to find ΔH_1

$q = mc\Delta T \mid \Delta H = -q/n$

Use graphical extrapolation for accurate ΔT

Apparatus

- Polystyrene cup with lid
- Thermometer (±0.1 °C) or data logger
- Measuring cylinder or pipette
- Stopwatch
- Reagents for Hess's law cycle (e.g., NaOH + HCl direct, and via NaOH + NH_4Cl then NH_3 + HCl)

Method

1. Measure a known volume of solution into the polystyrene cup.
2. Record temperature every minute for 3 minutes (baseline).
3. Add the second reagent at the 4th minute. Stir and record temperature for 6+ more minutes.
4. Plot temperature vs. time. Extrapolate to find the true ΔT.
5. Calculate $q = mc\Delta T$. Then $\Delta H = -q/n$.
6. **For Hess's Law:** Perform two separate experiments to measure two enthalpy changes in a cycle. Use Hess's Law to calculate the third (unmeasurable) enthalpy change.

Safety

- Acids and alkalis — corrosive, wear goggles.
- Exothermic reactions — cup may become warm.

Analysis

Plot: Temperature vs. time with graphical extrapolation.

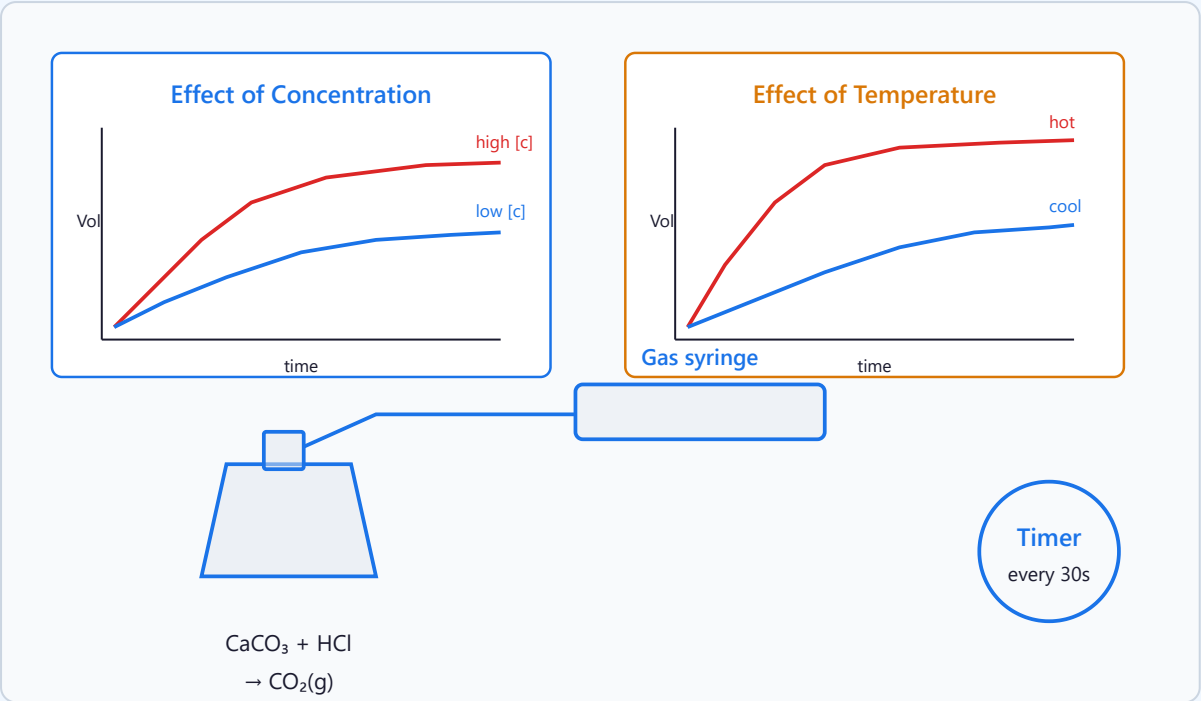
Hess's Law: $\Delta H_1 = \Delta H_2 + \Delta H_3$ (or appropriate cycle). Compare calculated value with literature.

Error Analysis

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic	Lid, insulation, graphical extrapolation
Assuming c = 4.18 and density = 1	Systematic	Acknowledged approximation
Heat absorbed by calorimeter	Systematic	Determine calorimeter constant
Incomplete mixing	Random	Stir thoroughly; use magnetic stirrer

Exam Tip: Hess's Law: enthalpy change for a reaction is independent of the route taken, provided initial and final conditions are the same. You can measure ΔH(formation) indirectly by measuring ΔH(combustion) of reactants and products. Draw a clear energy cycle in your answer. Always use graphical extrapolation of your temperature-time plot to correct for heat loss — this significantly improves accuracy.

⚗ PAG4: Rates of Reaction



Apparatus

- Conical flask, gas syringe, delivery tube, bung
- Stopwatch
- Measuring cylinders
- $CaCO_3$ (marble chips) and dilute HCl (for concentration investigation)
- $Na_2S_2O_3$ and HCl (for temperature investigation)
- Water baths at various temperatures

Method

1. **Continuous monitoring (concentration):** Set up flask + gas syringe. Use different concentrations of HCl (same excess $CaCO_3$). Record volume of CO_2 every 30s. Plot volume vs. time for each concentration.
2. **Effect of temperature:** Use $Na_2S_2O_3$ + HCl disappearing cross method at different temperatures. Record time for cross to disappear. Rate = 1/t.
3. Control variables: keep total volume constant (for concentration); keep volumes constant (for temperature).

Safety

- HCl is irritant — goggles.
- SO_2 from thiosulfate reaction — ventilated area.
- Hot water baths — care with burns.

Analysis

Concentration: Plot volume vs. time curves. Initial gradient = initial rate. Plot initial rate vs. [HCl] to find order.

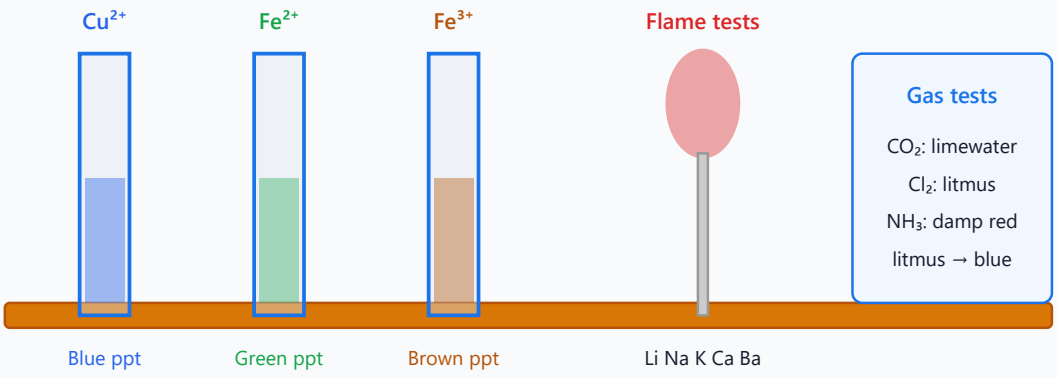
Temperature: Plot 1/t vs. T for exponential curve, or ln(1/t) vs. 1/T for Arrhenius plot. Gradient = $-E_a/R$.

Error Analysis

Source of Error	Type	Improvement
Gas escaping before bung fitted	Systematic	Seal quickly
Subjective cross disappearance	Random	Same observer; colorimeter
Temperature not maintained	Systematic	Use thermostatted water bath
Uneven marble chip sizes	Random	Use similar size/mass

Exam Tip: When drawing tangents on a volume-time curve, the tangent at t = 0 gives the initial rate. For the concentration investigation, if you use excess $CaCO_3$ with different [HCl], the initial rate changes but the final gas volume is the same (limited by the same mass of solid). For the disappearing cross experiment, rate = 1/t — this is an approximation of initial rate assuming the reaction is monitored over the same extent each time.

⚗ PAG5: Qualitative Tests for Ions



Anions: $Cl^-/Br^-/I^-$ ($AgNO_3$) | SO_4^{2-} ($BaCl_2$) | CO_3^{2-} (HCl + limewater)

Apparatus

- Test tubes, nichrome wire, Bunsen burner
- NaOH(aq), NH_3(aq), $AgNO_3$(aq), $BaCl_2$(aq)
- Dilute HNO_3, dilute HCl
- Limewater, litmus paper

Method

1. **Flame tests:** Clean wire in HCl, dip in sample, observe colour.
2. **Cation precipitate tests:** Add NaOH or NH_3 to sample. Record precipitate colour and behaviour in excess.
3. **Halide tests:** Acidify with HNO_3, add $AgNO_3$. White (Cl^-), cream (Br^-), yellow (I^-).
4. **Sulfate test:** Acidify with HNO_3, add $BaCl_2$. White ppt = SO_4^{2-}.
5. **Carbonate test:** Add HCl. Gas through limewater — milky = CO_3^{2-}.
6. **Ammonium test:** Add NaOH, warm gently. Test gas with damp red litmus — turns blue = NH_4^+.

Safety

- $AgNO_3$ stains — gloves. $BaCl_2$ is toxic.
- Conc. HCl in fume cupboard.

- Goggles throughout.

Analysis

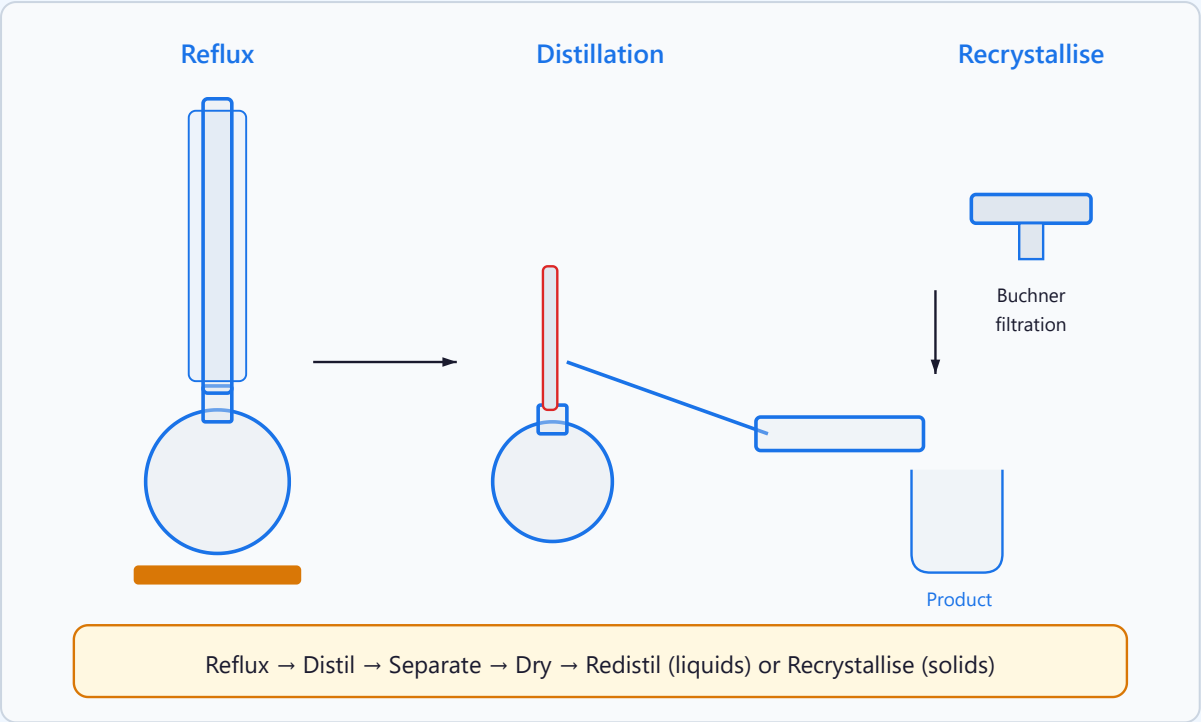
Record: Systematic table of observations for each test and each unknown ion.

Error Analysis

Source of Error	Type	Improvement
Contaminated wire/glassware	Systematic	Clean thoroughly before each test
Na^+ masking other flame colours	Systematic	View through blue cobalt glass
Not acidifying before $AgNO_3/BaCl_2$	Systematic	Always acidify with HNO_3 first
Similar precipitate colours	Random	Compare with known standards

Exam Tip: $Al(OH)_3$ dissolves in excess NaOH (amphoteric) but not in excess NH_3. $Cu(OH)_2$ dissolves in excess NH_3 (forms deep blue $[Cu(NH_3)_4]^{2+}$) but not in excess NaOH. This is the key distinguishing test in exams. Always acidify with dilute HNO_3 before adding $AgNO_3$ (halide test) or $BaCl_2$ (sulfate test) to prevent false positives from carbonates or sulfites.

⚗ PAG6: Organic Preparation and Purification



Apparatus

- Round-bottomed flask, anti-bumping granules
- Reflux condenser (vertical), distillation condenser (angled)
- Separating funnel
- Buchner funnel, filter flask, vacuum pump (for solids)
- Drying agent (anhydrous $MgSO_4$ or Na_2SO_4)
- Melting point apparatus (solids) or thermometer (liquids)

Method

1. **Reflux:** Heat reaction mixture with a vertical condenser to ensure complete reaction without losing volatile components.
2. **Distillation:** Rearrange for distillation. Collect product at the correct boiling point range.
3. **Liquid purification:** Use separating funnel to wash with Na_2CO_3 (removes acids) and water. Dry with anhydrous drying agent. Filter. Redistil.
4. **Solid purification (recrystallisation):** Dissolve in minimum hot solvent, hot filter, cool slowly, Buchner filter, wash with cold solvent, dry.

5. Test purity: melting point (solids) or boiling point (liquids).

Safety

- Flammable organics — no naked flames; electric heating.
- Fume cupboard for volatile substances.
- Never heat sealed apparatus.
- Vent separating funnel to release pressure.

Analysis

Calculate: Percentage yield = (actual/theoretical) × 100. Atom economy = (M_r of desired product / sum of M_r of all products) × 100.

Purity: Sharp melting/boiling point matching literature value = pure.

Error Analysis

Source of Error	Type	Improvement
Product lost during transfers	Systematic	Minimise transfers; rinse equipment
Incomplete reaction	Systematic	Reflux for longer; use excess reagent
Impure product (broad mp/bp)	Systematic	Repeat recrystallisation/distillation
Wet product (lowers mp)	Systematic	Dry thoroughly; use desiccator

Exam Tip: A pure solid has a sharp melting point matching the data book value. Impurities lower and broaden the melting point. For organic liquids, purity is confirmed by a sharp boiling point. Always explain each purification step and why it is necessary. Common yield-reducing factors: incomplete reaction, loss during transfer, loss during washing/filtration, side reactions.

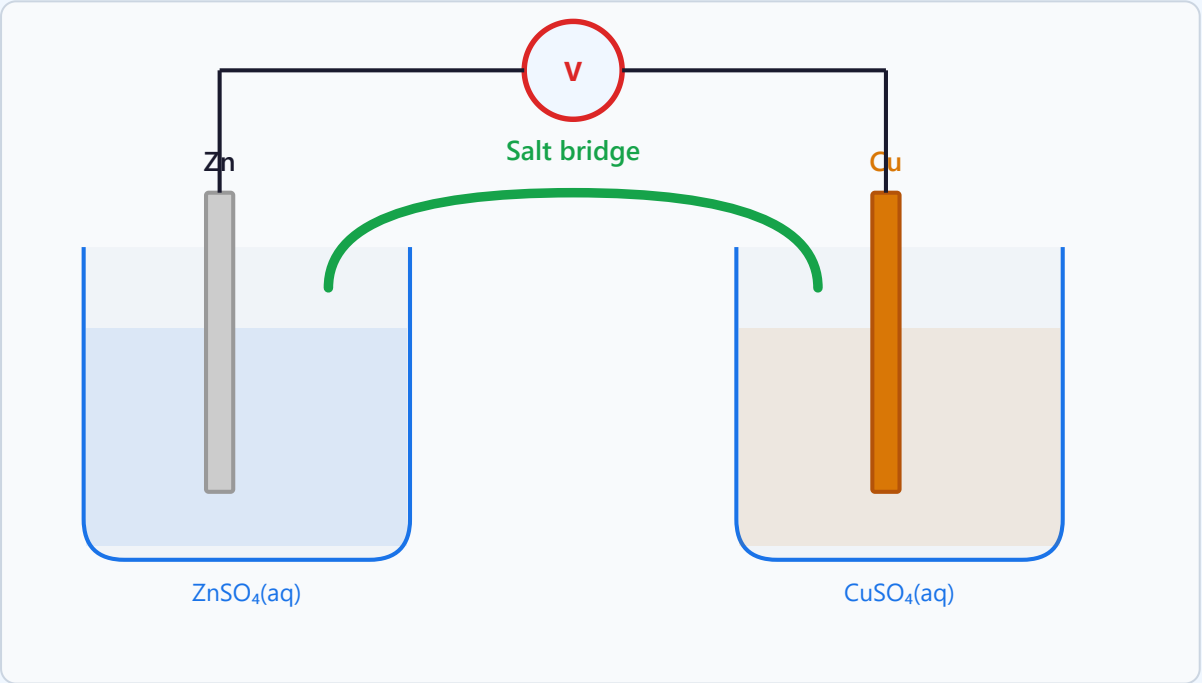
Worked Example — Percentage Yield:

A student prepared aspirin from 2.00 g of salicylic acid. The theoretical yield is 2.61 g. The actual yield after purification was 1.85 g.

Percentage yield = (1.85 / 2.61) × 100 = **70.9%**

This is a typical yield for a lab preparation. Losses occur during transfer, filtration, washing, and recrystallisation.

⚗ PAG7: Electrochemical Cells



Apparatus

- Beakers, metal electrodes (Zn, Cu, Fe, Mg, etc.)
- 1.0 mol dm^{-3} metal salt solutions
- Salt bridge (KNO_3), high-resistance voltmeter
- Connecting wires, crocodile clips, emery paper

Method

1. Clean electrodes with emery paper.
2. Set up half-cells in separate beakers with their metal ion solutions.
3. Connect with a salt bridge and wires to voltmeter.
4. Record EMF and polarity.
5. Investigate: different metal pairs, effect of concentration, effect of temperature.
6. Compare with E°(cell) = E°(cathode) - E°(anode).

Safety

- Metal salt solutions — wear gloves and goggles.
- Dispose of solutions correctly.

Analysis

Calculate: E°(cell) = E°(right) - E°(left). Compare measured and theoretical.

Investigate: How changing [ion] affects EMF (Nernst equation qualitatively).

Error Analysis

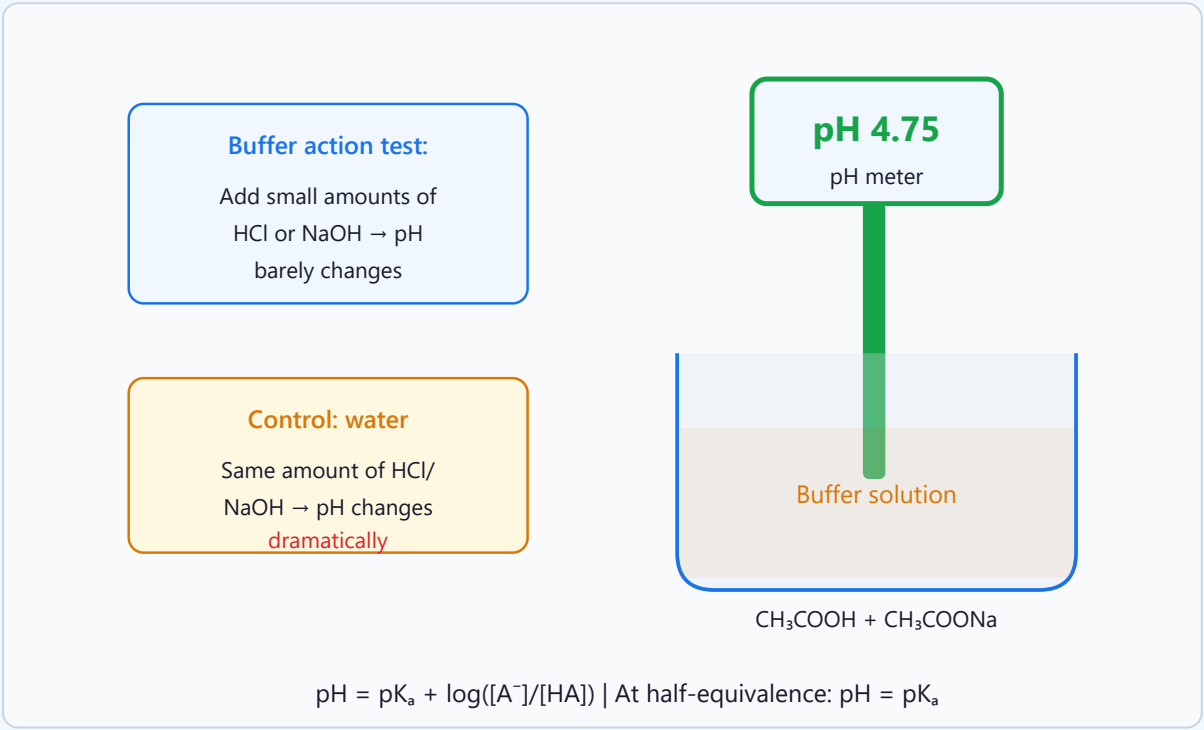
Source of Error	Type	Improvement
Oxide on electrodes	Systematic	Clean immediately before use
Non-standard conditions	Systematic	Record T; use 1.0 mol dm^{-3} solutions
Salt bridge resistance	Systematic	Fresh salt bridge; good contact
Voltmeter drawing current	Systematic	High-resistance voltmeter

Exam Tip: Electrons flow from the more negative electrode (anode, oxidation) to the more positive electrode (cathode, reduction) through the external circuit. Ions flow through the salt

bridge to maintain electrical neutrality. The salt bridge does not allow electron flow.

Remember: **AN OX** (anode = oxidation), **RED CAT** (reduction = cathode).

⚗ PAG8: pH Measurement and Buffer Solutions



Apparatus

- Calibrated pH meter or pH probe
- Buffer solutions: weak acid (e.g., CH_3COOH) + its conjugate base (CH_3COONa)
- Dilute HCl and NaOH (for testing buffer action)
- Measuring cylinders, beakers
- Distilled water (as control)

Method

1. Calibrate the pH meter using standard buffer solutions.

2. Prepare a buffer solution by mixing known volumes of weak acid and its sodium salt.
3. Measure the initial pH of the buffer.
4. Add a small measured volume of dilute HCl. Record the new pH.
5. Prepare a fresh buffer and add the same volume of NaOH. Record the pH.
6. Repeat steps 4-5 with distilled water as a control.
7. Compare: the buffer pH barely changes, while the water pH changes dramatically.
8. **Extension:** Prepare buffers with different ratios of $[HA]:[A^-]$ and verify that $pH = pK_a + \log([A^-]/[HA])$.

Safety

- Weak acids are irritants — goggles.
- NaOH is corrosive — gloves.
- pH probes are fragile — handle carefully.

Analysis

Calculate: Expected pH using Henderson-Hasselbalch: $pH = pK_a + \log([A^-]/[HA])$. Compare with measured pH.

Buffer capacity: Number of moles of H^+/OH^- needed to change pH by 1 unit.

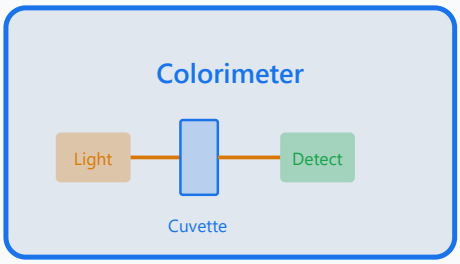
Error Analysis

Source of Error	Type	Improvement
pH meter not calibrated	Systematic	Calibrate with at least 2 standard buffers
Inaccurate volumes of HA and A^-	Random	Use pipettes for precise volumes
CO_2 from air dissolving in solutions	Systematic	Use freshly prepared solutions; work quickly
Temperature effects on pH	Systematic	Record temperature; work at constant T

Exam Tip: A buffer resists pH change because: when H^+ is added, A^- reacts with it ($A^- + H^+ \rightarrow HA$); when OH^- is added, HA reacts with it ($HA + OH^- \rightarrow A^- + H_2O$). The buffer works best

when [HA] ≈ [A^-] (i.e., pH ≈ pK_a). Maximum buffer capacity is at the half-equivalence point of a titration.

PAG9: Transition Metal Chemistry



Absorbance: 0.45

Transition Metal Complexes

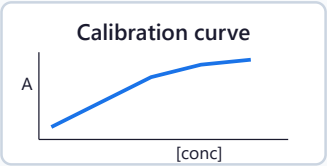
$[Cu(H_2O)_6]^{2+}$ — pale blue

$[Cu(NH_3)_4(H_2O)_2]^{2+}$ — deep blue

$[Cr(H_2O)_6]^{3+}$ — green

$[Cr(NH_3)_6]^{3+}$ — purple

$[Co(H_2O)_6]^{2+}$ — pink



$$A = \varepsilon c l$$

A = absorbance

ε = molar absorptivity

c = concentration, l = path length

Apparatus

- Colorimeter with appropriate colour filter, cuvettes
- Standard solutions of transition metal complex (various known concentrations)
- Test tubes for ligand substitution reactions
- Solutions: $CuSO_4$, NH_3(aq), NaOH(aq), HCl, EDTA

Method

1. **Colorimetry:** Prepare a series of standard solutions of known concentration.
2. Select the complementary colour filter on the colorimeter (e.g., red/orange filter for blue Cu^{2+} solutions).

3. Zero the colorimeter using distilled water as the blank.
4. Measure the absorbance of each standard. Plot a calibration curve: absorbance vs. concentration.
5. Measure the absorbance of the unknown solution. Read the concentration from the calibration graph.
6. **Ligand substitution:** Add NH_3 dropwise to $CuSO_4$(aq). Observe colour change from pale blue to deep blue as NH_3 replaces H_2O ligands. Record observations for each drop added.

Safety

- NH_3 has irritant fumes — work in well-ventilated area.
- Transition metal salts can be harmful — wear gloves.
- Handle cuvettes by the top (fingerprints affect readings).

Analysis

Plot: Calibration curve of absorbance vs. $[Cu^{2+}]$. Should be linear (Beer-Lambert Law: $A = \varepsilon cl$).

Calculate: Unknown concentration by interpolation from the calibration graph.

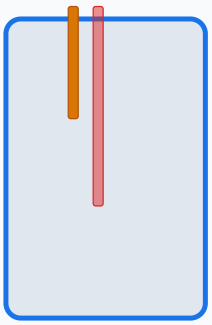
Error Analysis

Source of Error	Type	Improvement
Fingerprints on cuvette	Systematic	Handle by top; wipe with lens tissue
Wrong filter selected	Systematic	Use complementary colour filter; determine λmax
Air bubbles in cuvette	Random	Tap gently to dislodge; fill carefully
Dilution errors in standard solutions	Random	Use volumetric flasks and pipettes

Exam Tip: The filter used in the colorimeter must be the complementary colour to the solution being measured. A blue solution absorbs orange/red light, so use an orange filter.

The relationship A = εcl (Beer-Lambert Law) is only valid at low concentrations — at high concentrations, the calibration curve deviates from linearity. Always include a blank (distilled water) to zero the instrument.

⚗ PAG10: Organic Analysis



Melting point apparatus

Lab Tests

Br_2(aq) → C=C
2,4-DNP → C=O
Tollens' → CHO
Na_2CO_3 → COOH
$K_2Cr_2O_7/H^+$ → OH
Iodoform → $COCH_3$

Spectroscopy

Mass spec → M_r
IR → functional groups
(O-H, C=O, N-H etc.)
^{1}H NMR → H environments
^{13}C NMR → C environments
Splitting, integration, δ

Apparatus

- Melting point apparatus, capillary tubes
- Test tubes, water bath
- Reagents: Br_2(aq), 2,4-DNP, Tollens', Na_2CO_3, $K_2Cr_2O_7/H^+$, I_2/NaOH
- Spectroscopic data (mass spec, IR, NMR — provided in exams)

Method

1. Measure the melting point of the unknown solid. Compare with literature values.
2. Perform functional group tests systematically (see PAG5/RP6 methods).
3. Record all positive and negative results in a table.
4. Combine lab results with spectroscopic data to identify the compound:

- Mass spectrum: molecular ion peak gives M_r; fragmentation pattern gives structural clues.
- IR spectrum: broad O-H at 2500-3300 cm^{-1} (carboxylic acid), sharp C=O at ~1700 cm^{-1}, broad O-H at 3200-3550 cm^{-1} (alcohol).
- NMR: number of peaks = number of unique H/C environments. Integration ratio = relative number of H. Splitting pattern (n+1 rule).

Safety

- All standard organic test precautions apply.
- $K_2Cr_2O_7$ is carcinogenic — gloves, fume cupboard.
- Tollens' — prepare fresh, dispose immediately.

Analysis

Combine: Melting point, functional group tests, mass spec (M_r), IR (bonds), NMR (structure) to fully identify the compound.

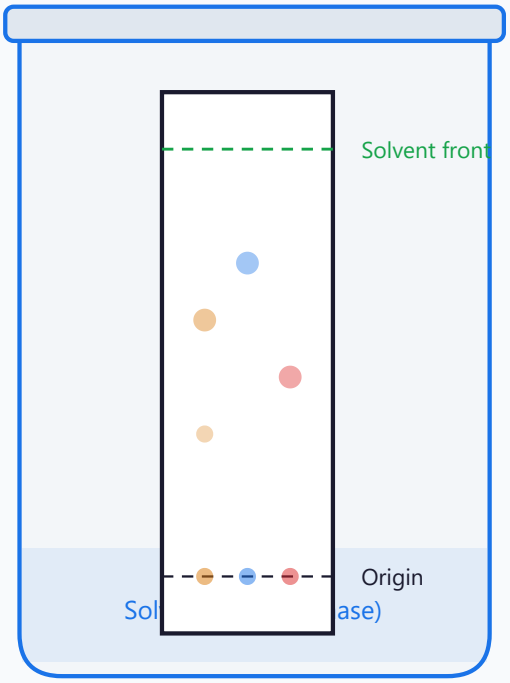
Error Analysis

Source of Error	Type	Improvement
Impure sample (broad mp)	Systematic	Purify first by recrystallisation
False positive/negative in tests	Systematic	Run positive and negative controls alongside
Heating too fast for melting point	Systematic	Heat slowly near expected mp
Contaminated NMR solvent	Systematic	Use pure deuterated solvent

Exam Tip: In combined analytical questions, start with the mass spectrum (molecular ion gives M_r and molecular formula), then check IR for key functional groups, and finally use NMR to determine the exact structure. The number of ^1H NMR peaks tells you the number of different hydrogen environments; integration gives the ratio; splitting (n+1 rule) tells you

adjacent hydrogens. Always calculate the degree of unsaturation (IHD) from the molecular formula — IHD = 4 strongly suggests a benzene ring.

PAG11: Chromatography



Rf Value

$$Rf = \frac{\text{distance by spot}}{\text{distance by solvent}}$$

Rf is constant for given solvent + substance

Key points:

Pencil line (not ink)
Solvent below origin
Lid prevents solvent evaporation

Apparatus

- TLC plates (silica or alumina on plastic/glass backing)
- Developing chamber (beaker with lid/cling film)
- Capillary tubes (for spotting), pencil, ruler
- Suitable solvent or solvent mixture (mobile phase)
- UV lamp or iodine vapour (for visualising colourless spots)

Method

1. Draw a faint pencil line (origin) about 1 cm from the bottom of the TLC plate.
2. Spot samples onto the origin using capillary tubes. Keep spots small and evenly spaced. Include known reference compounds.
3. Pour solvent into the developing chamber to a depth of about 0.5 cm (below the origin line). Allow atmosphere to saturate.
4. Place the plate in the chamber. Ensure the origin line is above the solvent level.
5. Cover and allow the solvent to rise up the plate by capillary action.
6. When the solvent front nears the top (~1 cm from edge), remove the plate and immediately mark the solvent front with pencil.
7. If spots are colourless, visualise under UV lamp or in iodine vapour.
8. Measure distances from origin to centre of each spot and to the solvent front.
9. Calculate Rf = distance by spot / distance by solvent front.

Safety

- Many solvents are flammable and volatile — no flames; fume cupboard.
- UV lamp — do not look directly at the UV light.
- Iodine vapour is irritant — use in fume cupboard.

Analysis

Calculate: Rf values for each spot. Compare with reference values or known standards run alongside.

Identify: Components with the same Rf as known compounds (in the same solvent system) are likely the same substance.

Error Analysis

Source of Error	Type	Improvement
Spots too large (poor resolution)	Systematic	Apply small, concentrated spots; dry between applications

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Solvent level above origin line	Systematic	Check depth before inserting plate
Solvent front not marked immediately	Random	Mark immediately on removal — solvent evaporates quickly
Chamber not saturated with vapour	Systematic	Line chamber with filter paper; allow time to saturate
Using ink pen instead of pencil	Systematic	Always use pencil — ink will dissolve in the solvent

Exam Tip: R^f values are only valid if compared under identical conditions (same solvent, temperature, plate type). A substance is identified if its R^f matches a known reference run alongside it in the same experiment. Two spots from an unknown sample at the origin means the unknown contains at least two components. TLC is a form of adsorption chromatography — polar substances adsorb more strongly to the polar silica stationary phase and have lower R^f values. Non-polar substances travel further (higher R^f).

⚗ PAG12: Investigate Equilibria

Le Chatelier's Principle

Kc Determination

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

Titrate with NaOH to find [acid] at equilibrium

Kc = [ester][water] / [acid][alcohol]

$[Co(H_2O)_6]^{2+}$(pink) + $4Cl^-$ ⇌ $[CoCl_4]^{2-}$(blue) + $6H_2O$ ΔH = +ve

Only temperature changes Kc. Concentration shifts position but not Kc.

Apparatus

- Test tubes, water baths (ice and hot)
- $CoCl_2$ solution, conc. HCl, distilled water
- **For Kc:** Sealed tubes with ethanoic acid + ethanol + catalyst (H_2SO_4), NaOH for titration, burette, pipette, indicator

Method

1. **Le Chatelier's (qualitative):** Prepare cobalt equilibrium mixture (purple). Add HCl → blue. Add water → pink. Heat → blue. Cool → pink.
2. **Kc determination (quantitative):** Seal mixtures of ethanoic acid and ethanol with catalyst in tubes. Leave at constant temperature until equilibrium (several days or heat to speed up).
3. Titrate the equilibrium mixture with standardised NaOH to find moles of acid remaining.
4. Use ICE table: calculate moles of each species at equilibrium.
5. Calculate Kc = [products]/[reactants].

Safety

- $CoCl_2$ is toxic — gloves.
- Conc. HCl — fume cupboard.

- Conc. H_2SO_4 catalyst — very corrosive.

Analysis

Qualitative: Record colour changes and link to Le Chatelier's principle.

Quantitative: ICE table to calculate equilibrium concentrations. Kc = $[CH_3COOC_2H_5][H_2O]$ / $[CH_3COOH][C_2H_5OH]$.

Error Analysis

Source of Error	Type	Improvement
Equilibrium not reached	Systematic	Allow sufficient time; use catalyst; check with repeat
Temperature not constant	Systematic	Use thermostatted water bath
Titrating disturbs equilibrium	Systematic	Quick titration; acknowledged limitation
Catalyst (H_2SO_4) also titrated by NaOH	Systematic	Run a blank with just H_2SO_4 to subtract; account for in calculation

Exam Tip: For the esterification $K^{)}$: remember to subtract the moles of H_2SO_4 catalyst from the titre (it also reacts with NaOH). The value of $K^{)}$ for this reaction is approximately 4 at room temperature. A catalyst does not change $K^{)}$ or the equilibrium position — it only speeds up attainment of equilibrium. When describing Le Chatelier's shifts, always state: (1) what change was made, (2) which direction the equilibrium shifts, (3) why (to counteract the change), and (4) what observable effect this has (colour change).

Worked Example — $K^{)}$ from Esterification:

1.00 mol ethanoic acid and 1.00 mol ethanol were sealed with H_2SO_4 catalyst. At equilibrium, titration with NaOH showed 0.333 mol acid remaining (after subtracting the catalyst contribution).

Moles reacted = 1.00 − 0.333 = 0.667 mol. From stoichiometry, moles of ester = 0.667, moles of water = 0.667, moles of alcohol = 1.00 − 0.667 = 0.333.

$K^{)}$ = (0.667 × 0.667) / (0.333 × 0.333) = 0.445 / 0.111 = **4.01** (no units — moles cancel)